

# SMART INDIA HACKATHON 2026



## CRYPTAGUARD



**SMART INDIA  
HACKATHON  
2026**

**Problem Statement ID – SIH26164**

**Problem Statement Title – “Enterprise Cryptographic  
Discovery & Analysis Tool (ECDAT)”**

**Theme – Blockchain & Cybersecurity**

**PS Category – Software**

**Team ID – 138525**

**Team Name (Registered on portal) – Cypher Vault**

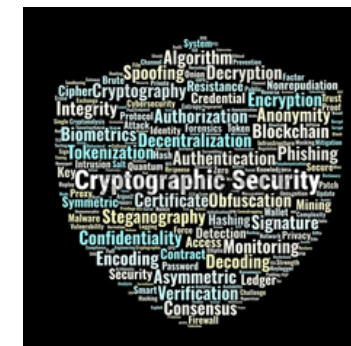
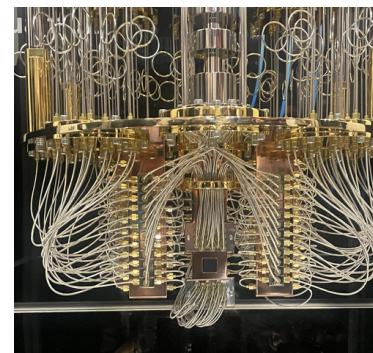
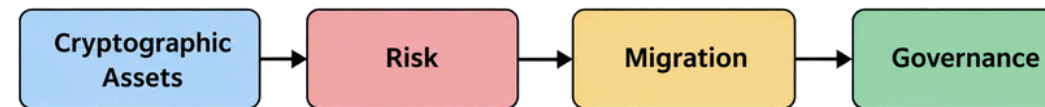


## The Cryptographic Challenge

- Cryptographic assets can be found within the source code, libraries, certificates, configuration, binaries, and containers that modern businesses use in the organization.
- With quantum computing posing a threat to traditional public key cryptography, the organizations need to understand which of the assets are vulnerable and exposed.

## From Discovery to Decision

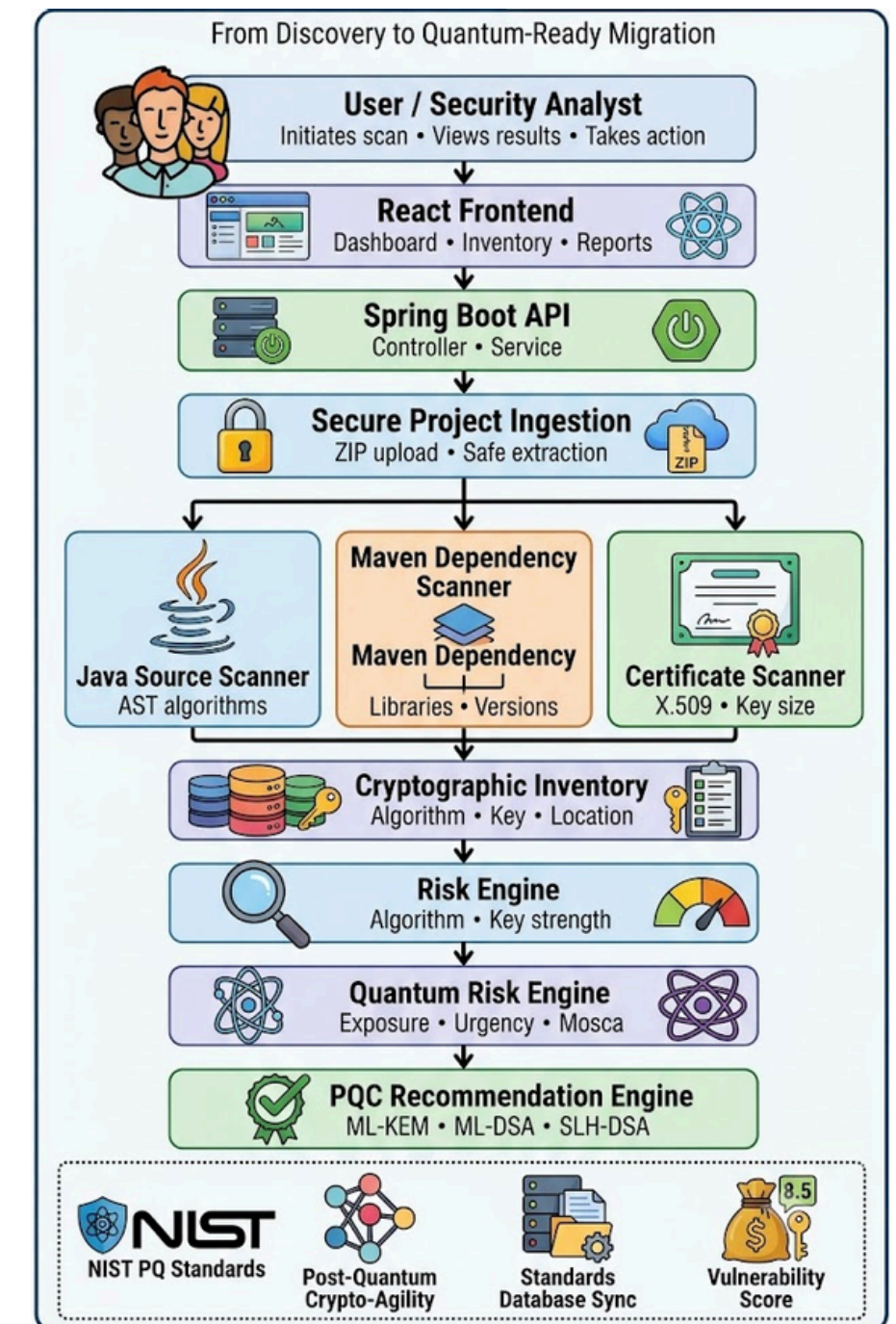
- Discover what cryptography exists.
- Understand what is at risk.
- Prioritize what needs to change.
- Identify the path toward PQC.



## CRYPTAGUARD

- CRYPTAGUARD delivers an end-to-end cryptographic discovery to migration decision framework.
- It conducts cryptographic discovery and inventorying, analysis using classical and quantum risk, priority migration based on cryptographic life time and Mosca analysis, and suggests appropriate PQC / hybrid solutions.
- All above deliverables connect in a migration roadmap and CBOM.

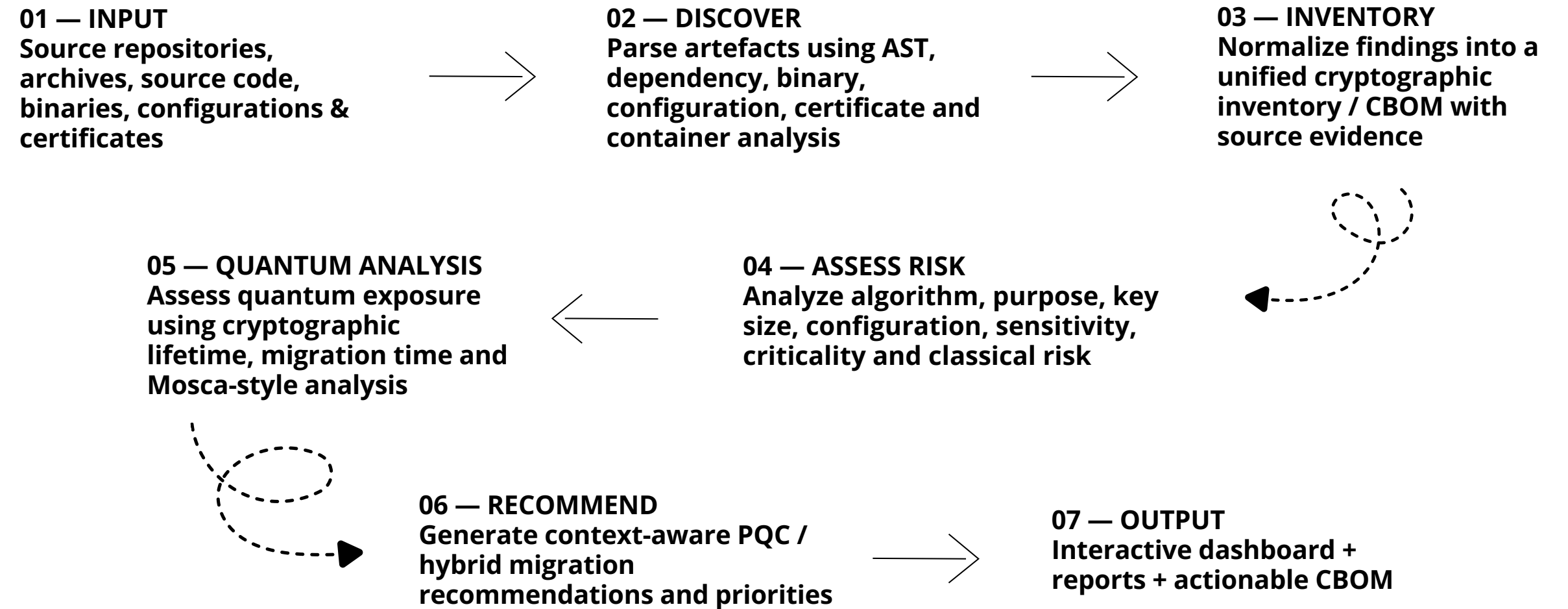
## ARCHITECTURE



## TECHNOLOGY STACK



## METHODOLOGY



"The CRYPTAGUARD prototype works, allowing cryptographic discovery, inventory management, risk assessment against quantum attacks, PQC/hybrid recommendations, migration prioritization, and generation of CBOMs."

## Technical Feasibility

End-to-end prototype including discovery, inventory, risk analysis and migration advice.

Multi-method analysis based on source code, dependencies, configurations, certificates, binaries and containers.

Conclusion is supported by cryptographic use, location of source code and contextual risk data.

Flexible design allowing other scanning techniques, cryptographic policies and migration advice to be added.

## Competitor Check

**Crypto Scanners** → detect cryptography, limited migration context

**SBOM Platforms** → track components, limited crypto/quantum analysis

**PQC Tools** → recommend algorithms, limited asset discovery

## Operational Viability

Actionable workflow based on cryptographic discovery → quantum risk assessment → migration prioritization.

CBOM-enabled governance links cryptographic assets to risk, recommendation and migration.

Dashboard for interactive analysis, visualization and reporting of security teams.

Foundation for future enterprise integrations and extended cryptographic asset discovery.

| Challenge           | Risk                                   | Mitigation                        |
|---------------------|----------------------------------------|-----------------------------------|
| Crypto Discovery    | Fragmented & hidden assets             | Multi-modal discovery + inventory |
| Quantum Threat      | Exposure of vulnerable cryptography    | Quantum + Mosca-based analysis    |
| Mitigation Priority | Difficult to identify what comes first | Risk-based prioritization         |
| PQC Selection       | Unclear migration alternatives         | PQC / Hybrid recommendations      |

# IMPACT AND BENEFITS

- **Security & Quantum Readiness:**

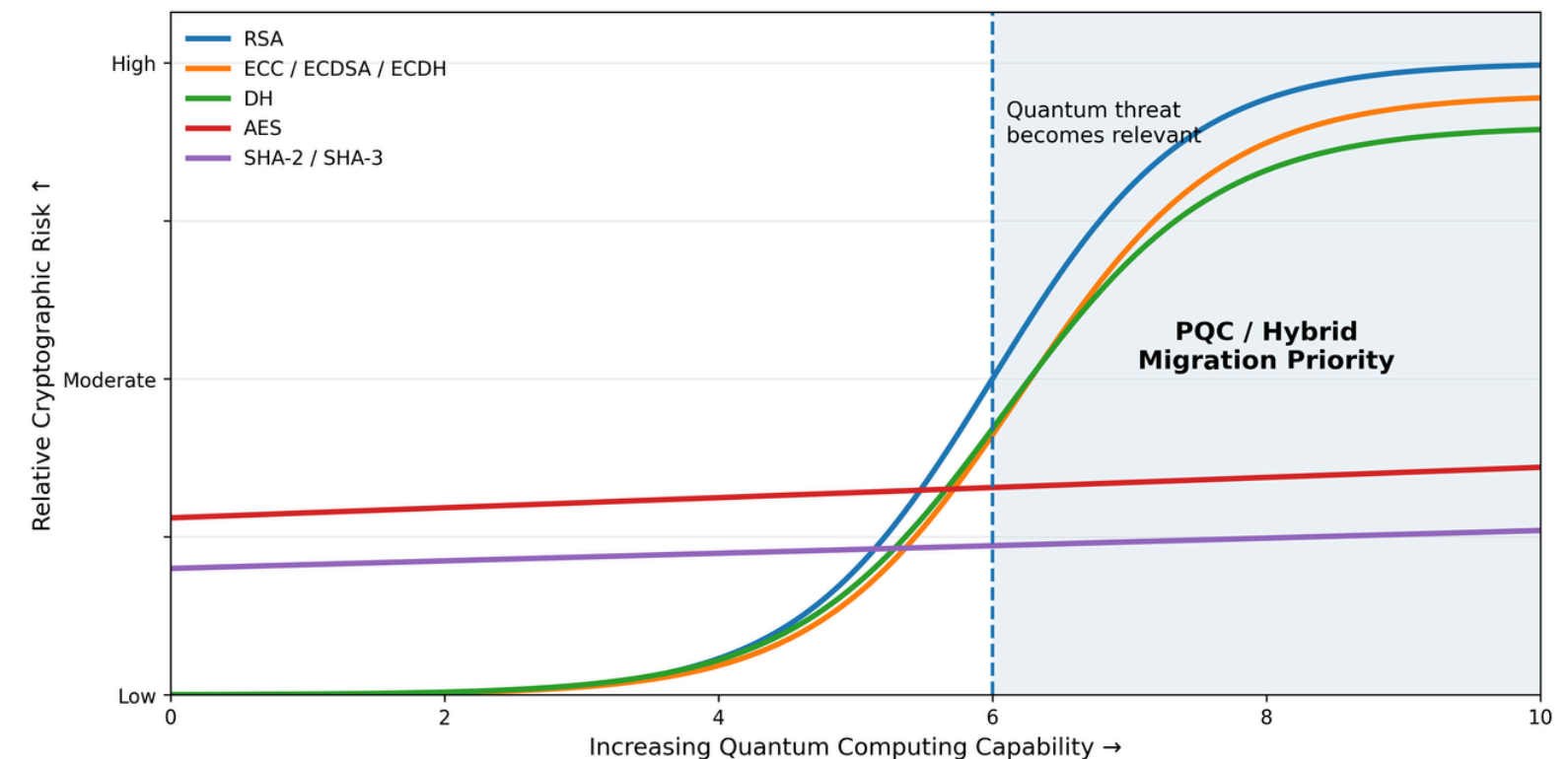
Ensures visibility for cryptography assets and quantum-vulnerable usage, aiding organizations in understanding the areas that require migration.

- **Operational & Migration Efficiency**

This transforms disparate pieces of information related to cryptography into a migration roadmap, allowing security professionals to focus on assets that need early evaluation and migration.

- **Governance & Compliance Visibility**

Creates a centralized cryptographic inventory and actionable CBOM, connecting assets, evidence, risk and recommended migration paths for security governance.



Conceptual illustration; not a forecast of when a cryptographically relevant quantum computer will arrive.

**“As quantum capability advances, CRYPTAGUARD helps organizations move from cryptographic visibility to prioritized migration readiness.”**

## RESEARCH FOUNDATION

- Cryptographic discovery and inventory are foundational for PQC migration, enabling organizations to identify where and how cryptography is used across their systems.
  - Quantum-vulnerable public-key cryptography requires planned migration toward quantum-resistant alternatives, supported by risk assessment and prioritization.
- CRYPTAGUARD extends this approach by connecting discovery and inventory with quantum-risk analysis, migration prioritization and PQC/hybrid recommendations.

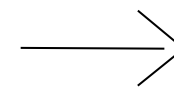
Primary Reference: NIST SP 1800-38B — Cryptographic Discovery **NIST SP 1800-38B**

## SOLUTIONS ALREADY EXIST

**IBM Quantum Safe Explorer** — *Cryptographic discovery, CBOM generation and migration prioritization.*

**IBM Guardium Cryptography Manager** — *Enterprise cryptographic inventory, risk assessment and lifecycle management.*

**Keyfactor AgileSec** — *Cryptographic asset discovery across code, infrastructure, networks and cloud.*



## OUR SOLUTION STANDS OUT

- **Unified workflow:** *Discovery → Inventory → Risk → Migration → CBOM*
- **Multi-modal analysis:** *source code, dependencies, configurations, certificates, binaries and containers*
- **Quantum-focused assessment:** *combines cryptographic exposure, lifetime and Mosca-based analysis*

*You cannot secure what you cannot see — CRYPTAGUARD makes your cryptographic estate visible, measurable, and actionable.*